# COMPUTER NETWORKS

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| Ex.No.14 | **To Analyze the different types of servers using Webalizer tool** |
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**Aim**

To **install, configure, and execute the Webalizer log analysis tool on a Windows operating system** to analyze raw server logs and generate interactive HTML dashboards containing traffic graphs and usage statistics.

**Apparatus Required**

- **Operating System:** Windows 10 or Windows 11 (64-bit)
- **Software Bundle:** XAMPP for Windows (includes Apache Web Server and pre-configured Webalizer)
- **Access Level:** Windows Administrator privileges
- **Web Browser:** Microsoft Edge, Google Chrome, or Mozilla Firefox

**Theory**

Server log files record every user transaction, but they are difficult to read in their raw .log text formats. **Webalizer** solves this by converting text logs into structured HTML pages.

While Webalizer is natively a Linux tool, it is seamlessly integrated into Windows environments using **XAMPP**. XAMPP automatically pipes the Apache web server log (access.log) into the Webalizer engine. Webalizer then calculates key performance indicators:

1. **Hits:** Total number of files requested (images, text, pages).
2. **Files:** Total number of successful requests fulfilled by the server.
3. **Pages:** Pure HTML or PHP document views (ignoring individual embedded images).
4. **Visits:** Distinct user journeys tracked via unique IP addresses.

**Step-by-Step Procedure**

**Step 1: Install the XAMPP Server Environment**

1. Download the XAMPP installer for Windows from the official website.

2. Double-click the installer executable file.
3. During the **Select Components** prompt, ensure that **Apache** and **Webalizer** are checked.
4. Complete the wizard to install XAMPP into the default directory: C:\xampp\.

**Step 2: Start the Web Server**

1. Open the Windows Start Menu, search for **XAMPP Control Panel**, and run it as an **Administrator**.
2. Click the **Start** button next to the **Apache** module. The module background will turn green, indicating the web server is actively running and generating log files.

**Step 3: Locate the Log and Configuration Files**

Verify where Windows is storing your logs and how Webalizer is targeting them:

- **Raw Server Logs Location:** C:\xampp\apache\logs\access.log
- **Webalizer Configuration File Location:** C:\xampp\webalizer\webalizer.conf

*(Optional Tuning)*: If you open webalizer.conf using Notepad, you will observe that XAMPP has pre-written the paths using forward-slash formatting compatible with the tool:

```
text
LogFile c:/xampp/apache/logs/access.log
OutputDir c:/xampp/htdocs/webalizer
Use code with caution.
```

**Step 4: Execute Webalizer Manually via Windows Command Prompt**

1. Press Win + R, type cmd, and hit Enter to launch the **Command Prompt**.
2. Change the directory to the XAMPP Webalizer folder:
   ```
   cmd
   cd C:\xampp\webalizer
   Use code with caution.
   ```
3. Run the executable tool to process the active Windows logs:
   ```
   cmd
   webalizer.exe
   Use code with caution.
   ```
   *The command prompt will flash an information line confirming how many lines of text data it successfully read and converted.*

**Step 5: View and Analyze the HTML Reports**

1. Launch your web browser.
2. Type the following address into the URL bar and press enter:
   http://localhost/webalizer/
3. Click on the current calendar month link to view the visual traffic analysis charts, total system bandwidth usage, and data tables tracking user visits.

**Step 6: Automate the Log Analysis using Windows Task Scheduler**

To avoid running the command prompt manually every day, set up a background Windows Task:

1. Press the Windows key, search for **Task Scheduler**, and open it.
2. In the right-hand Actions menu, select **Create Basic Task**.
3. Name your task XAMPP Webalizer Update and select **Daily** as the frequency trigger.
4. Set the Action to **Start a program**.

5. Click **Browse** and target the executable path: C:\xampp\webalizer\webalizer.exe
6. In the **Start in (optional)** textbox, type C:\xampp\webalizer to ensure the program finds its default configuration parameters. Click **Finish**.

## Conclusion

The Webalizer analysis program was successfully deployed and executed on the Windows OS environment via XAMPP. Raw, unorganized data entries in access.log were converted into comprehensive, easy-to-read web graphs. The setup provides administrators with a highly visual framework to evaluate server stress, incoming traffic spikes, and user distribution patterns.